

Hybrid Tangent Spaces

Beyond Smooth Dynamics: Impacts, Switches, and Discontinuities

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2026-01-18

Abstract

The Tangent Hyperplane framework, as developed in the core thesis, assumes C^1 smoothness of the vector field—a requirement that excludes a vast class of physically important systems. Robotic foot strikes, ball impacts, friction cone transitions, and mode-switching controllers all exhibit velocity discontinuities or derivative discontinuities that violate the smoothness assumption. This article extends the tangent space framework to hybrid dynamical systems, where dynamics evolve continuously within modes but jump discontinuously across guard surfaces. We develop the geometric machinery of tangent spaces at boundaries, introduce saltation matrices that map perturbations across jumps, and derive mode-aware trajectory optimization algorithms. The result is a unified framework encompassing both smooth and non-smooth dynamics, with rigorous treatment of the measure-theoretic subtleties that arise when differentiating through discontinuities.

Part I: Motivation and Theory

Chapter 1: Why Smoothness Fails

The C^1 Assumption and Its Limits

The entire Tangent Hyperplane framework rests on a single critical assumption:

! Smoothness Requirement (Main Thesis)

The vector field $f : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ is **continuously differentiable** (C^1), meaning:

$$\dot{x} = f(x, u), \quad f \in C^1 \quad (1)$$

This guarantees:

1. The Jacobian $A(x, u) = \frac{\partial f}{\partial x}$ exists and is continuous
2. Tangent spaces $T_x \mathcal{M}$ are well-defined at every state x
3. Linearized dynamics $\delta \dot{x} = A \delta x + B \delta u$ are exact in the infinitesimal limit

This assumption is **physically reasonable** for most classical mechanical systems derived from Lagrangian or Hamiltonian mechanics. Pendulums, robotic arms, spacecraft, and chemical reactors all produce smooth vector fields.

But consider the following scenarios:

! Where Smoothness Breaks Down

Example 1: Humanoid Foot Strike

A bipedal robot's foot collides with the ground. Before contact: $v_{foot} < 0$ (downward velocity). After contact: $v_{foot} = 0$ (instantaneous stop). The velocity experiences a **discontinuous jump** $\Delta v \neq 0$ over zero time.

At the instant of impact, the derivative $\dot{x}$ is undefined. The tangent space does not exist.

Example 2: Golf Ball Impact

A golf club head traveling at $v_{club} = 50$ m/s strikes a stationary ball. The collision lasts $\Delta t \approx 0.5$ ms. During this interval, contact forces $F \sim 10^4$ N produce accelerations $\ddot{x} \sim 10^6$ m/s². The force model $F(x, \dot{x})$ is discontinuous at the contact surface.

The Jacobian $\frac{\partial f}{\partial x}$ has a singularity at $x_{contact}$.

Example 3: Coulomb Friction

The Coulomb friction model is:

$$f_{friction} = \begin{cases} -\mu_s F_N \cdot \text{sgn}(\dot{x}) & \text{if } |\dot{x}| > 0 \\ -F_{applied} & \text{if } |\dot{x}| = 0, |F_{applied}| < \mu_s F_N \\ -\mu_k F_N \cdot \text{sgn}(\dot{x}) & \text{if sliding} \end{cases} \quad (2)$$

The signum function $\text{sgn}(\dot{x})$ is discontinuous at $\dot{x} = 0$. The derivative $\frac{\partial f}{\partial \dot{x}}$ does not exist at stiction transitions.

Example 4: Bang-Bang Control

Fuel-optimal control for thrust-limited spacecraft often produces discontinuous control policies:

$$u^*(x) = \begin{cases} u_{max} & \text{if } \lambda^T B > 0 \\ 0 & \text{if } \lambda^T B = 0 \\ -u_{max} & \text{if } \lambda^T B < 0 \end{cases} \quad (3)$$

where λ is the costate from Pontryagin's Maximum Principle. The control $u(t)$ switches discontinuously, making $f(x, u(x))$ non-smooth even if f itself is smooth in its arguments.

These are not pathological edge cases—they are **core phenomena** in robotics, aerospace, and mechanical engineering. A framework that cannot handle them is incomplete.

The Failure of Naïve Linearization

What happens if we blindly apply the tangent space framework to a system with discontinuities? Consider a simple 1D impact model:

$$\dot{v} = -g, \quad v^+ = -e \cdot v^- \quad \text{when } x = 0 \quad (4)$$

where $e \in [0, 1]$ is the coefficient of restitution. A ball falls under gravity until it hits the ground ($x = 0$), at which point the velocity reverses with energy loss.

Attempt 1: Ignore the discontinuity

If we linearize around a trajectory that includes an impact, computing $\delta\dot{x} = A\delta x$ at the impact instant gives:

$$A_{\text{impact}} = \frac{\partial}{\partial v} (-e \cdot v^-) = \text{undefined}$$

The derivative does not exist because the velocity is a function of the left-limit v^- , not the state itself. The Jacobian is **singular**.

Attempt 2: Treat impact as a stiff force

We might approximate the impact as a very stiff spring:

$$F_{\text{contact}}(x) = \begin{cases} 0 & x > 0 \\ -kx - c\dot{x} & x \leq 0 \end{cases}, \quad k \rightarrow \infty$$

This makes f continuous but introduces **numerical stiffness**. Explicit integrators require timesteps $\Delta t \sim 1/\sqrt{k}$, which becomes prohibitively small. Implicit integrators work but lose the analytical structure we seek.

Attempt 3: Smooth via mollification

Replace the discontinuous jump with a smooth transition:

$$v^+ = -e \cdot v^- \cdot \sigma_\epsilon(x)$$

where $\sigma_\epsilon(x) = \frac{1}{2}(1 + \tanh(x/\epsilon))$ is a sigmoid. This makes $f \in C^\infty$, but:

1. **Accuracy depends on ϵ :** Too large and physics is wrong, too small and we're back to stiffness
2. **Loss of structure:** The sharp transition at $x = 0$ has physical meaning (guard surface), which is obscured
3. **Parameter tuning:** Every discontinuity requires choosing ϵ , with no principled method

The Need for Hybrid Formalism

None of these workarounds is satisfactory. What we need is a **mathematically rigorous framework** that:

1. **Accepts discontinuities** as first-class features, not approximations
2. **Defines tangent spaces** on either side of jumps, even if not at the jump itself
3. **Maps perturbations** across discontinuities via a well-defined linear operator
4. **Integrates into optimization**, preserving the computational advantages of DDP/iLQR

This framework already exists: **hybrid dynamical systems**. The goal of this article is to extend the tangent space perspective to hybrid systems, developing the geometric tools needed to handle impacts, switches, and discontinuities rigorously.

Chapter Summary

The C^1 smoothness assumption is not a minor technicality—it is **central** to the Tangent Hyperplane framework. Without it:

- Jacobians may not exist
- Tangent spaces are undefined
- Linearization fails

Physically important systems (impacts, friction, mode switches) violate this assumption. Naïve workarounds (ignoring, stiffening, mollifying) all have fatal flaws.

The solution is to **extend the framework** to hybrid systems, where discontinuities are handled explicitly via guard surfaces and jump maps. This is the subject of the remainder of this article.

Chapter 2: Hybrid Automata Primer

What is a Hybrid System?

A **hybrid dynamical system** is one that exhibits both continuous evolution and discrete jumps. Mathematically, it consists of:

 **Definition:** Hybrid Automaton

A hybrid automaton $\mathcal{H}$ is a tuple:

$$\mathcal{H} = (Q, \mathcal{X}, \mathcal{U}, F, G, R, \Sigma) \quad (5)$$

where:

- $Q = \{q_1, q_2, \dots, q_m\}$ is a finite set of **discrete modes** (also called locations or charts)
- $\mathcal{X} = \mathbb{R}^n$ is the **continuous state space**
- $\mathcal{U} = \mathbb{R}^p$ is the **control input space**
- $F = \{f_1, f_2, \dots, f_m\}$ is a family of **vector fields**, one per mode:

$$\dot{x} = f_q(x, u), \quad q \in Q \quad (6)$$

- $G = \{G_{ij} \subseteq \mathcal{X} \mid i, j \in Q\}$ is a family of **guard sets** (switching surfaces)
- $R = \{R_{ij} : G_{ij} \rightarrow \mathcal{X} \mid i, j \in Q\}$ is a family of **reset maps** (jump maps)
- $\Sigma = \{e_{ij} \mid i, j \in Q\}$ is the set of discrete **events** (mode transitions)

Intuition: A hybrid system is like a finite state machine where each state has its own continuous dynamics. While in mode q , the state evolves according to $\dot{x} = f_q(x, u)$. When x enters a guard set G_{ij} , an event e_{ij} is triggered, the state jumps via the reset map $x^+ = R_{ij}(x^-)$, and the mode switches from q_i to q_j .

Example: Bouncing Ball

The bouncing ball from Equation ?? can be formalized as a hybrid automaton:

Discrete modes:

$$Q = \{\text{flight}\}$$

(Single mode—ball is always in flight when not at boundary)

Continuous state:

$$x = \begin{bmatrix} h \\ v \end{bmatrix}, \quad h = \text{height}, \quad v = \text{velocity}$$

Vector field (flight mode):

$$f_{\text{flight}}(x) = \begin{bmatrix} v \\ -g \end{bmatrix} \quad (7)$$

Guard set:

$$G = \{x \in \mathbb{R}^2 \mid h = 0, v < 0\} \quad (8)$$

The ball triggers the guard when it reaches the ground ($h = 0$) while moving downward ($v < 0$).

Reset map:

$$R(x^-) = \begin{bmatrix} 0 \\ -e \cdot v^- \end{bmatrix} \quad (9)$$

The height remains zero, but the velocity reverses and scales by the restitution coefficient e .

Execution: The system evolves as:

1. Integrate $\dot{x} = f_{\text{flight}}(x)$ continuously
2. Monitor guard condition $h = 0, v < 0$
3. When guard is triggered:
 - Compute pre-impact state x^-
 - Apply reset: $x^+ = R(x^-)$
 - Reinitialize continuous evolution from x^+
4. Repeat

This formalism makes the discontinuity **explicit** and **localized** to the guard surface.

Multi-Mode Example: Walking Robot

A more realistic example is a simplified walking robot with stance and swing phases:

Discrete modes:

$$Q = \{\text{stance}, \text{swing}\}$$

Continuous state:

$$x = \begin{bmatrix} \theta \\ \phi \\ \dot{\theta} \\ \dot{\phi} \end{bmatrix}$$

where θ = torso angle, ϕ = leg angle.

Vector fields:

Stance mode (foot on ground, single support):

$$M_s(x)\ddot{q} + C_s(x, \dot{q})\dot{q} + G_s(x) = B_s u + J_s^T \lambda \quad (10)$$

where λ is the contact force (computed via constraint).

Swing mode (foot in air, no contact):

$$M_{sw}(x)\ddot{q} + C_{sw}(x, \dot{q})\dot{q} + G_{sw}(x) = B_{sw} u \quad (11)$$

Guards:

$G_{\text{stance} \rightarrow \text{swing}}$: Toe-off condition, e.g., $\lambda_N = 0$ (normal force vanishes)

$G_{\text{swing} \rightarrow \text{stance}}$: Heel-strike condition, e.g., $h_{\text{foot}} = 0$ (foot contacts ground)

Reset maps:

At heel strike, the foot velocity must become zero (instantaneous impact):

$$R_{\text{swing} \rightarrow \text{stance}}(x^-) = \begin{bmatrix} q^- \\ \Phi(q^-)\dot{q}^- \end{bmatrix} \quad (12)$$

where $\Phi(q)$ is the **impact map** derived from collision equations (see Section ??).

At toe-off, state is continuous:

$$R_{\text{stance} \rightarrow \text{swing}}(x^-) = x^- \quad (13)$$

Filippov Solutions at Discontinuities

What happens if the guard surface is not a single isolated event but a region where the system “slides” along a discontinuity? This occurs in:

- Coulomb friction at $\dot{x} = 0$ (stiction)
- Saturation limits where $u = u_{max}$ for extended time
- Complementarity constraints in contact mechanics

The mathematical tool for defining solutions is the **Filippov convex hull**:

i Filippov Solutions

At a discontinuity surface $S = \{x \mid h(x) = 0\}$ where the vector field is multi-valued, define the Filippov set-valued map:

$$F_{\text{Filippov}}(x) = \bigcap_{\delta > 0} \bigcap_{\mu(N)=0} \overline{\text{conv}} f(B(x, \delta) \setminus N) \quad (14)$$

In words: The Filippov convex hull is the convex closure of all limit values of f in arbitrarily small neighborhoods, ignoring sets of measure zero.

For piecewise smooth systems with two modes separated by $h(x) = 0$:

$$F_{\text{Filippov}}(x) = \begin{cases} \{f_1(x)\} & h(x) > 0 \\ \{f_2(x)\} & h(x) < 0 \\ \text{conv}\{f_1(x), f_2(x)\} & h(x) = 0, \nabla h \cdot f_1 \cdot \nabla h \cdot f_2 < 0 \end{cases} \quad (15)$$

The last case (sliding mode) occurs when the vector fields on either side point toward the surface.

Example: Coulomb friction with $\dot{x} = 0$:

If the applied force is less than the maximum static friction, the system remains at $\dot{x} = 0$ via a Filippov solution. The friction force $f \in [-\mu_s N, \mu_s N]$ takes whatever value is needed to maintain $\ddot{x} = 0$.

Relevance to tangent spaces: Filippov solutions are the measure-theoretic generalization of classical solutions. They allow us to define trajectories through discontinuities in a way compatible with Lebesgue integration. This is essential for variational analysis, as we'll see in Section ??.

Measure-Theoretic Interpretation

Why do we need measure theory? Because **derivatives are defined via limits**, and limits require a notion of “almost everywhere” convergence.

In a hybrid system, the trajectory $x(t)$ is:

- **Absolutely continuous** within each mode (standard ODE solution)
- **Discontinuous** at guard crossings (jumps in x)
- **Piecewise smooth** overall

The derivative $\dot{x}(t)$ exists **almost everywhere** (i.e., everywhere except the finite or countable set of jump times). This is sufficient for:

- Calculus of variations (action integrals $\int L dt$ are well-defined)

- Necessary conditions for optimality (Pontryagin's Maximum Principle)
- Numerical integration (Lebesgue integral equals Riemann integral a.e.)

Key insight: We do not need $\dot{x}$ to exist **everywhere**—only almost everywhere (a.e.). Jumps have measure zero, so they contribute zero to integrals:

$$\int_0^T L(x(t), u(t)) dt = \sum_{i=1}^{N-1} \int_{t_i}^{t_{i+1}} L(x(t), u(t)) dt \quad (16)$$

where $t_1, \dots, t_{N-1}$ are the jump times (finite set).

Chapter Summary

Hybrid systems formalize discontinuous dynamics via:

1. **Modes** (Q): Discrete states with their own continuous dynamics
2. **Guards** (G): Surfaces triggering transitions
3. **Resets** (R): Jump maps applied at transitions

Examples include bouncing balls (single mode, impulsive reset) and walking robots (multi-mode with foot strikes).

Filippov solutions extend classical ODE solutions to discontinuous vector fields via convex hulls, enabling sliding modes.

Measure theory ensures derivatives exist almost everywhere, making variational calculus and optimization well-defined despite jumps.

Next, we develop the geometry of tangent spaces at jumps, enabling linearization across discontinuities.

Chapter 3: Tangent Spaces at Jumps

Left and Right Tangent Spaces

At a jump, the state $x(t)$ is discontinuous. Limits from the left and right differ:

$$x^- = \lim_{t \rightarrow t_j^-} x(t), \quad x^+ = \lim_{t \rightarrow t_j^+} x(t), \quad x^+ = R(x^-) \quad (17)$$

The critical question: **How do perturbations δx behave across the jump?**

In the smooth case, perturbations evolve via the state transition:

$$\delta x(t) = \Phi(t, t_0) \delta x(t_0) \quad (18)$$

where Φ satisfies $\frac{\partial}{\partial t} \Phi(t, t_0) = A(t) \Phi(t, t_0)$.

In the hybrid case, we need a **generalized state transition** that accounts for jumps. Define:

 **Definition: Left and Right Tangent Spaces at a Jump**

At a guard crossing at time t_j :

Left tangent space: $T_{x^-} \mathcal{M}^- = \mathbb{R}^n$ (standard tangent space before jump)

Right tangent space: $T_{x^+} \mathcal{M}^+ = \mathbb{R}^n$ (standard tangent space after jump)

Jump map on tangent spaces: The reset map $R : G_{ij} \rightarrow \mathcal{X}$ induces a linear map on tangent spaces:

$$\boxed{DR(x^-) : T_{x^-} G_{ij} \rightarrow T_{x^+} \mathcal{X}} \quad (19)$$

This is the **Jacobian of the reset map**, denoted:

$$P_j = \left. \frac{\partial R}{\partial x} \right|_{x^-} \quad (20)$$

Perturbations jump according to:

$$\delta x^+ = P_j \delta x^- \quad (21)$$

Geometric interpretation: The reset map R is a smooth function from the guard surface G_{ij} to the state space $\mathcal{X}$. Its derivative DR is a linear map between tangent spaces. Even though the trajectory itself jumps discontinuously, **perturbations evolve linearly** according to P_j .

Jump Map Jacobians

Let's compute P_j for our examples.

Example 1: Bouncing Ball

The reset map is:

$$R \begin{pmatrix} h^- \\ v^- \end{pmatrix} = \begin{pmatrix} 0 \\ -ev^- \end{pmatrix}$$

Taking the Jacobian:

$$P_j = \frac{\partial R}{\partial x} = \begin{bmatrix} \frac{\partial h^+}{\partial h^-} & \frac{\partial h^+}{\partial v^-} \\ \frac{\partial v^+}{\partial h^-} & \frac{\partial v^+}{\partial v^-} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & -e \end{bmatrix} \quad (22)$$

Interpretation:

- A perturbation in height δh^- does not affect the post-impact state (first column is zero)
- A perturbation in velocity δv^- scales by $-e$ (second column)
- The matrix is **rank-deficient** (rank 1), reflecting the constraint $h^+ = 0$

Example 2: Rigid Body Impact (Poisson's Hypothesis)

For a rigid body colliding with a surface, conservation of momentum during impact gives:

$$Mv^+ = Mv^- + J^T \Lambda \quad (23)$$

where J is the contact Jacobian and Λ is the impulsive force. Using the restitution law:

$$Jv^+ = -eJv^-$$

Solving for v^+ :

$$v^+ = v^- - (1 + e)M^{-1}J^T(JM^{-1}J^T)^{-1}Jv^- \quad (24)$$

The Jacobian is:

$$P_j = I - (1 + e)M^{-1}J^T(JM^{-1}J^T)^{-1}J \quad (25)$$

This is the **impact map Jacobian**, a projection matrix that removes and reverses the normal component of velocity.

Saltation Matrices (Anitescu & Potra)

The jump map Jacobian P_j is part of a more general structure called the **saltation matrix**, which accounts for not just the jump in state but also the **variation in jump time**.

! Saltation Matrix

Consider a trajectory that crosses a guard surface $S = \{x \mid h(x) = 0\}$ at time t_j . If we perturb the initial condition $x_0 \rightarrow x_0 + \delta x_0$, both the jump time and the jump state vary:

$$t_j \rightarrow t_j + \delta t_j, \quad x^- \rightarrow x^- + \delta x^-$$

The **saltation matrix** S_j maps the pre-jump perturbation (including time variation) to the post-jump perturbation:

$$\delta x^+ = S_j \delta x^- - f^+ \delta t_j \quad (26)$$

where:

$$S_j = P_j + \frac{(f^+ - P_j f^-) \nabla h^T}{\nabla h^T f^-} \quad (27)$$

and:

- $P_j = \frac{\partial R}{\partial x} \big|_{x^-}$ is the reset Jacobian
- $f^- = f(x^-, u^-)$, $f^+ = f(x^+, u^+)$ are the vector fields before/after
- ∇h is the gradient of the guard function
- $\delta t_j = \frac{\nabla h^T \delta x^-}{\nabla h^T f^-}$ is the variation in crossing time

Derivation (sketch):

The guard crossing condition is $h(x(t_j)) = 0$. Under a perturbation:

$$h(x^- + \delta x^-) = 0 \implies \nabla h^T \delta x^- + O(\delta x^2) = 0$$

But δx^- includes evolution up to time $t_j + \delta t_j$:

$$\delta x^-(t_j + \delta t_j) = \Phi(t_j, t_0) \delta x_0 + \Phi(t_j, t_0) f^- \delta t_j$$

The crossing time variation is:

$$\nabla h^T (\delta x^- + f^- \delta t_j) = 0 \implies \delta t_j = -\frac{\nabla h^T \delta x^-}{\nabla h^T f^-}$$

After the jump:

$$\delta x^+ = P_j \delta x^- + (\text{correction for time variation})$$

The correction term comes from the fact that we reinitialize at x^+ at time $t_j + \delta t_j$, not t_j . Combining these:

$$\delta x^+ = P_j \delta x^- + (f^+ - P_j f^-) \delta t_j$$

Substituting δt_j and simplifying yields Equation ??.

Physical interpretation:

The saltation matrix has two contributions:

1. P_j : The direct effect of the reset map on the state
2. **Correction term**: Accounts for the fact that perturbed trajectories cross the guard at different times, experiencing different amounts of pre- and post-impact flow

Example: Bouncing Ball Saltation Matrix

For the bouncing ball:

- $h(x) = h$ (height), so $\nabla h = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$
- $f^- = \begin{bmatrix} v^- \\ -g \end{bmatrix}$, $f^+ = \begin{bmatrix} -ev^- \\ -g \end{bmatrix}$
- $P_j = \begin{bmatrix} 0 & 0 \\ 0 & -e \end{bmatrix}$

Computing the correction:

$$f^+ - P_j f^- = \begin{bmatrix} -ev^- \\ -g \end{bmatrix} - \begin{bmatrix} 0 \\ -ev^- - eg \end{bmatrix} = \begin{bmatrix} -ev^- \\ (1-e)(-g) + ev^- \end{bmatrix}$$

Wait, let me recalculate:

$$P_j f^- = \begin{bmatrix} 0 & 0 \\ 0 & -e \end{bmatrix} \begin{bmatrix} v^- \\ -g \end{bmatrix} = \begin{bmatrix} 0 \\ eg \end{bmatrix}$$

$$f^+ - P_j f^- = \begin{bmatrix} -ev^- \\ -g \end{bmatrix} - \begin{bmatrix} 0 \\ eg \end{bmatrix} = \begin{bmatrix} -ev^- \\ -g(1+e) \end{bmatrix}$$

$$\nabla h^T f^- = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} v^- \\ -g \end{bmatrix} = v^- < 0$$

The saltation matrix is:

$$\begin{aligned} S_j &= \begin{bmatrix} 0 & 0 \\ 0 & -e \end{bmatrix} + \frac{1}{v^-} \begin{bmatrix} -ev^- \\ -g(1+e) \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & -e \end{bmatrix} + \begin{bmatrix} -e & 0 \\ -g(1+e)/v^- & 0 \end{bmatrix} \\ &\boxed{S_j = \begin{bmatrix} -e & 0 \\ -\frac{g(1+e)}{v^-} & -e \end{bmatrix}} \end{aligned} \tag{28}$$

Interpretation:

- First row: Height perturbation δh^+ depends on pre-impact velocity perturbation δv^- via time-of-crossing variation
- Second row: Velocity perturbation jumps by $-e$, with additional dependence on height perturbation due to gravity during the time correction

State Transition Through Multiple Jumps

For a hybrid trajectory with multiple jumps at times $t_1, \dots, t_N$, the state transition operator is:

$$\boxed{\Phi(t, t_0) = \Phi_c(t, t_N) S_N \Phi_c(t_N, t_{N-1}) S_{N-1} \cdots S_1 \Phi_c(t_1, t_0)} \tag{29}$$

where:

- $\Phi_c(t_j, t_{j-1})$ is the continuous state transition within mode j (standard smooth case)
- S_j is the saltation matrix at jump j

This product structure allows us to **propagate perturbations** through arbitrarily complex hybrid trajectories, alternating between continuous flow and discrete jumps.

Chapter Summary

At guard crossings, states jump discontinuously, but **perturbations evolve linearly** via:

1. **Reset Jacobian** $P_j = \frac{\partial R}{\partial x}$: Direct effect of jump map
2. **Saltation Matrix** S_j : Includes correction for variation in crossing time
3. **Hybrid state transition** Φ : Product of continuous flows and saltation matrices

The key insight: **Even for non-smooth trajectories, the tangent space dynamics remain linear.** This extends the core Tangent Hyperplane principle to hybrid systems.

Next, we develop the geometric interpretation of these jumps as transformations of manifolds.

Part II: Geometric Framework

Chapter 4: Manifolds with Boundaries

Guard Surfaces as Codimension-1 Manifolds

In smooth dynamics, the state space $\mathcal{X} = \mathbb{R}^n$ is an n -dimensional manifold without boundary. Trajectories flow smoothly through this space, never encountering edges or discontinuities.

In hybrid dynamics, guard surfaces partition the state space into regions. Each guard G_{ij} is a **codimension-1 submanifold**, meaning it has dimension $n - 1$ and locally divides $\mathbb{R}^n$ into two half-spaces.

💡 Definition: Guard Surface as Manifold

A guard surface G is the zero level set of a smooth function $h : \mathbb{R}^n \rightarrow \mathbb{R}$:

$$G = \{x \in \mathbb{R}^n \mid h(x) = 0\} \quad (30)$$

Assume h is C^1 and **regular**, meaning $\nabla h(x) \neq 0$ for all $x \in G$. By the **Implicit Function Theorem**, G is a smooth $(n - 1)$ -dimensional embedded submanifold.

Tangent space of G :

At $x \in G$, the tangent space $T_x G$ is the null space of the gradient: